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Early-warning of unsafe hoisting operations: An integration of digital twin and knowledge graph

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ABSTRACT

Unsafe hoisting operations have been consistently associated with numerous safety incidents involving tower cranes. Currently, the predominant measures to mitigate these operations center around comprehensive training and education, emphasizing standardized protocols prior to hoisting activities. Despite concerted efforts in this direction, a conspicuous research gap persists in early-warning mechanisms during the construction phase. This paper aims to address this gap by proposing an innovative early-warning methodology, inspired by the principles of digital twin and knowledge graph. We firstly introduce a digital twin framework designed to mirror the real-time operational status of the tower crane. This framework enables the immediate detection of deviations or infractions as they occur. Subsequently, we develop a knowledge graph capable of promptly identifying unsafe hoisting operations by leveraging real-time data obtained from the digital twin. To validate the efficacy of our proposed methodology, we construct a scaled-down replica of a tower crane and establish a tailored digital twin system. The findings of a series of experimental trials prominently underscore the system's capability to generate timely alerts in response to unsafe hoisting operations while maintaining an impressively low rate of false alarms.

1. Introduction

The construction industry has long acknowledged the considerable risks inherent in tower crane operations. Numerous academic studies and industrial reports have demonstrated the potential compromise of crane stability due to unsafe hoisting operations including untested lifting, aggressive operation, and excessive swinging (Wiethorn, 2018). While tower crane operators generally receive trainings regarding operational specifications, violations are inevitable during the actual construction process. Therefore, proactive intervention is necessary to guarantee operational safety.

Previous research has introduced proactive hoisting monitoring methods that can leverage Internet of Things (IoT) techniques to address issues like excessive loads, blind spots, and collisions (Zhu et al., 2023). Some studies have amalgamated data from various sensors to prevent intrusion events, particularly unauthorized entry into the hoisting operation area (Gu et al., 2012). Nevertheless, these studies often overlook the operational roles of tower crane operators, and do not extensively monitor various violations that may occur during the entire hoisting process.

Recent studies have increasingly adopted digital twin (DT) to capitalize on the immersive experience for hoisting safety. The DT can simulate realistic hoisting operations before actual construction, facilitating technical disclosure (Sepasgozar, 2020), remote training (Li et al., 2012), and other activities. Additionally, it provides comprehensive information to improve the navigation of hoisting routes (Ji and Leite, 2018). Studies have also incorporated finite element mechanical models into the DT of the tower crane to assess stability during hoisting operations (Jiang et al., 2022). An essential rationale for these applications is that the DT represents the correspondence between virtual and physical spaces in construction activities (Su et al., 2023) (Opoku et al., 2021) (Kang and Mo, 2024). The utilization of DT can be categorized into six levels, including simulation, mapping, control, prediction, optimization, and symbiosis (Tao et al., 2022). Existing applications of DT in tower crane operations primarily concentrate on simulation and mapping scenarios, resulting in underutilization of DT for controlling event occurrences.

To leverage the DT for event control, it is crucial to employ numerical models, rule-based reasoning, and artificial intelligence techniques that incorporate domain knowledge for event identification, fault

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Table 1
Example of unsafe hoisting operations according to relevant standards.

No.	Content
USA: ANSI/ASME B30.3–2019/**Text content:** Qualifications and Operation	
1	The operator shall test the brakes each time a load approaching the rated load is handled, by lifting it a few inches and applying the brakes
2	When swinging the boom or jib, trolleying a load, or traveling the crane, sudden starts and stops shall be avoided
3	The hook is brought over the load in such a manner as to minimize swinging
UK: BS-7121-5-2019/**Text content:** Safety and Operation Considerations	
1	The load should initially be lifted just clear of the supporting surface and brought to rest while the slings, balance of the load, etc., are checked before proceeding
2	Care should be exercised by the operator at all times to avoid: Shock or side loadings on the jib or structure
3	The hoisting, rotation, traversing, luffing or traveling motions of a crane should not be used to drag any load along the ground with the hoist rope out of the vertical position
CN: JGJ196-2010/**Text content:** Operation of Tower Cranes	
1	When the lifting load reaches 90% of the rated load, the object should be lifted 200 mm–500 mm above the ground, and all conditions should be checked before the lifting
2	The controller should be operated from zero gear, push into first gear, and then step by step increase the gear
3	The tower crane should not operate reverse rotation when the rotation is not stopped
ISO 12480-3-2020/**Text content:** Operation and Control	
1	The hoist rope, or if applicable the hoist chain, shall be vertical throughout the hoisting operation
2	The load shall initially be lifted just clear of the supporting surface and be brought to rest while the slings, balance of the load, etc. Are checked, before proceeding
3	Proper care shall be exercised by the driver at all times to avoid shock or side loadings on the jib or structure
4	Motion motors should not be reversed before the motor has come to rest

diagnosis, and automated decision-making (Tuhaise et al., 2023) (Sacks et al., 2020). Among these, the knowledge graph serves as structured representations of real-world knowledge (Ji et al., 2022), which can effectively enhance knowledge representation, interoperability, and reasoning within the DT (Karabulut et al., 2024). Different knowledge graphs have been established for subway and building construction (Xing et al., 2019) (Gao et al., 2022). However, limited research has employed the knowledge graph for identifying unsafe hoisting operations, thereby constraining the potential for predictive control within the DT.

To address these concerns, this study aims to develop an early-warning methodology for controlling unsafe tower crane hoisting operations by integrating DT and knowledge graph. An extensive literature review is conducted to analyze common unsafe hoisting operations. Subsequently, a DT framework for hoisting operations is introduced, within which, the DT not only offers real-time mapping of the actual hoisting status in the physical world but also proactively alerts against unsafe hoisting operations. Additionally, a knowledge graph for identifying unsafe hoisting operations is introduced. To validate the methodology, we construct a scaled model of the tower crane along with its associated DT system, enabling multifaceted experiments on unsafe hoisting operations. Given the innovative application of DT and knowledge graph for early warning of unsafe hoisting operations, the results can significantly contribute to the existing body of knowledge.

The remainder of the paper is organized as follows. Section 2 introduces the related literature with a focus on the classification of unsafe tower crane operations and the management of hoisting safety. Section 3 presents the proposed early-warning methodology. Section 4 describes the evaluation of the proposed methodology with a scaled model. Section 5 presents the experimental results and discussions. Section 6 concludes this study.

2. Related works

2.1. Unsafe tower crane operations

Worker operations play a pivotal role in determining the safety risks associated with tower crane operations. In-depth safety analysis by Im and Park (2020) focused on crane safety regulations in South Korea, meticulously examining 288 safety provisions. Their findings highlighted the need for enhanced monitoring of worker operations. Chen et al. (2022) suggested that the optimal management strategy for tower crane operators involves stringent supervision, comprehensive maintenance practices, and strict adherence to safety regulations. Moreover, the Crane Safety Index identified the operator proficiency as a crucial factor shaping the safety landscape of crane operations (Mohandes et al., 2022). Lingard et al. (2021) formulated a model to comprehend the root causes of crane safety events, qualitatively analyzing lifting-related accidents and identifying a total of 77 contributing factors. Among these factors, several are intertwined with worker behaviors, such as deviations from instructions, lack of familiarity with machinery, and non-compliance with procedures. Similarly, case studies conducted in Hong Kong (Tam and Fung, 2011) and South Korea (Shin, 2015) underscored the prominent role of worker operations in on-site safety management.

On a global scale, various countries and regions have established standards that incorporate safety protocols for hoisting operations. Categorizing unsafe hoisting operations within these regulatory frameworks can be achieved through three principles. Firstly, by restricting the extraction of operational specifications to instances involving the practical use of tower cranes, excluding organizational management, installation, removal, and maintenance processes. Secondly, by focusing on operation specifications relevant only to tower crane operators, rather than other workers at the construction site. Thirdly, by scrutinizing operation specifications directly linked to the activity itself, without encompassing workers' physiological, psychological, or training aspects. Table 1 presents examples of unsafe hoisting operations from standards in the United States (The American Society of Mechanical Engineers, 2020), the United Kingdom (The British Standards Institution, 2019), China (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2010), and ISO (International Organization for Standardization, 2020).

2.2. Management of hoisting safety

The focal point of scholarly inquiry has consistently revolved around the management of hoisting safety (Sadeghi et al., 2021). Substantial scholarly contributions have focused on the development of intelligent methods aimed at enhancing hoisting activities. These methods encompass supporting operators during hoisting procedures and providing timely alerts for safety concerns.

In the realm of assisting hoisting operations, Shapira et al. (2014) introduced a laser scanning-based method to deal with blind spots. This approach effectively quantified both visible and concealed regions within the field of vision for tower crane operators. Lee et al. (2012) pioneered a tower crane navigation system that provided crane operators with comprehensive insights into their surroundings and the lifted object. Fang et al. (2018) proposed real-time reconstruction of the environment using computer vision technology, Inertial Measurement Unit (IMU), and scene modeling, enabling dynamic hoisting path planning to address safety risks posed by blind spots. Chi et al. (2012) studied on the user interface of remote tower crane operation systems, seeking to leverage comprehensive scene information for making informed operational decisions. Zhou et al. (2019) devised a swift-deploying, versatile IoT monitoring system, with a specific emphasis on large structure. Zhang et al. (2023) proposed immersive collaborative training for diverse hoisting tasks. Sun et al. (2023) introduced methods for collecting human/team behavior data and conducting quantitative risk

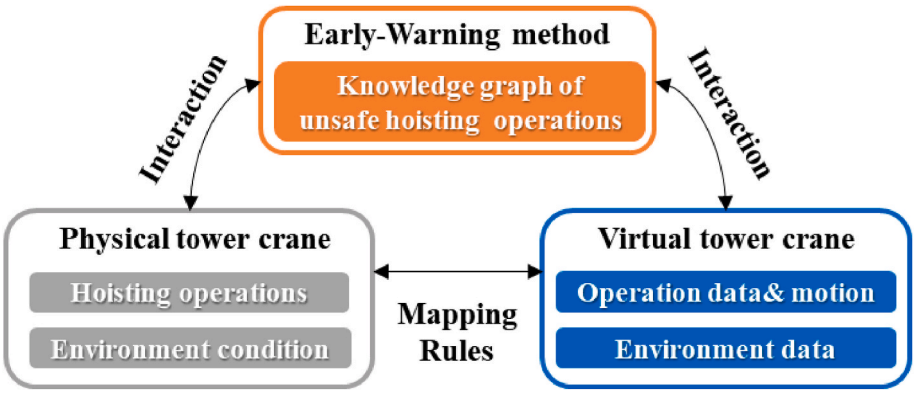


Fig. 1. DT framework for tower crane hoisting.

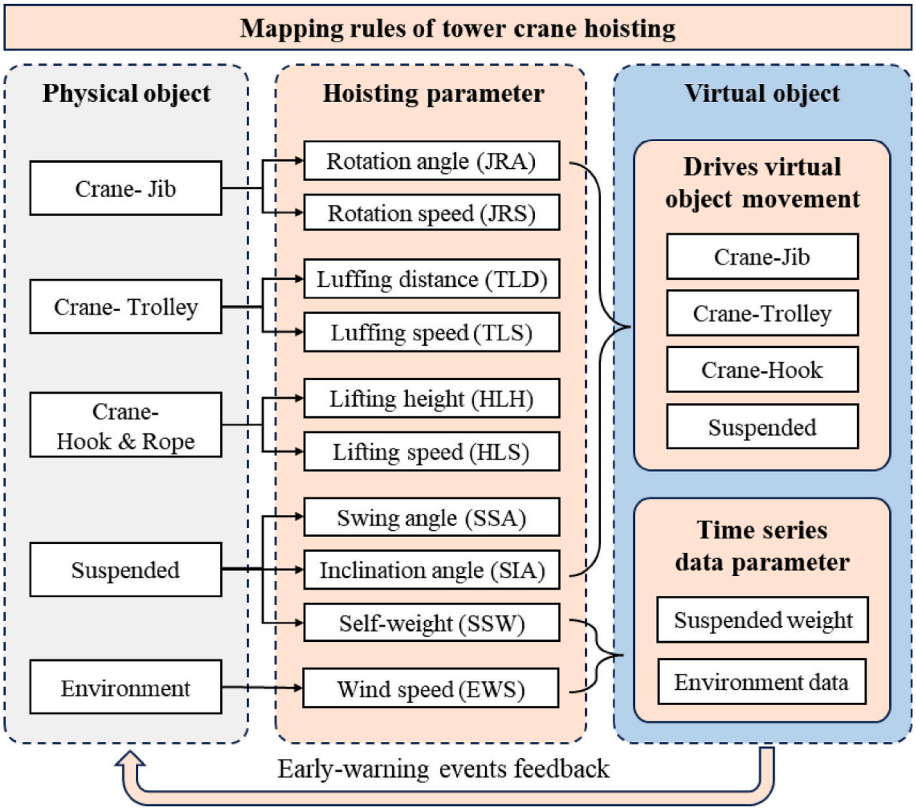


Fig. 2. Mapping rules of tower crane hoisting.

analyzes within the human-computer interaction system. The proposed strategies were rooted in the information physical system to ensure effective behavior control. Zhu et al. (2022) developed a lifting path planning tool, facilitating tower crane operations via heuristic algorithms and virtual prototyping. Cho and Han (2022) demonstrated hoisting operation planning in a 3D virtual space, leveraging augmented learning algorithms.

In terms of early-warning systems for hoisting safety risks. Yang et al. (2019) introduced a computer vision-based approach to extract the positions of workers and lifted objects, utilizing pixel-level data to calculate safety distances. Luo et al. (2011) presented an intelligent monitoring method to manage operation zones of tower cranes. Yang et al. (2014) introduced a method for monitoring tower crane operations via cameras, which can effectively discern activities such as concrete pouring and component delivery. Lalik et al. (2017) proposed dual measurement systems for monitoring stress variations in tower crane masts and jibs. Chian et al. (2022) investigated the risk of inadvertent unloading during tower crane operations and introduced a vision-based technique to dynamically identify the immediate areas below the load as the potential fall zones. Additionally, researchers have developed virtual systems from a systematic perspective to support hoisting operations and safety management. For instance, Kan et al. (2018) proposed a Cyber-Physical System-based modeling method for crawler crane operations, which can reduce collision risks by mapping spatial information. Similarly, Liu et al. (2021) developed a DT model for prefabricated building hoisting operations, enabling the analysis of various operational factors and their impacts on overall safety risks.

In essence, the primary thrust of research has focused on addressing several key issues, including rectifying blind spots inherent in hoisting operations, mitigating collisions involving tower cranes and nearby obstacles, and enhancing the management of access to operation zones. Despite these advancements, a notable gap remains concerning the

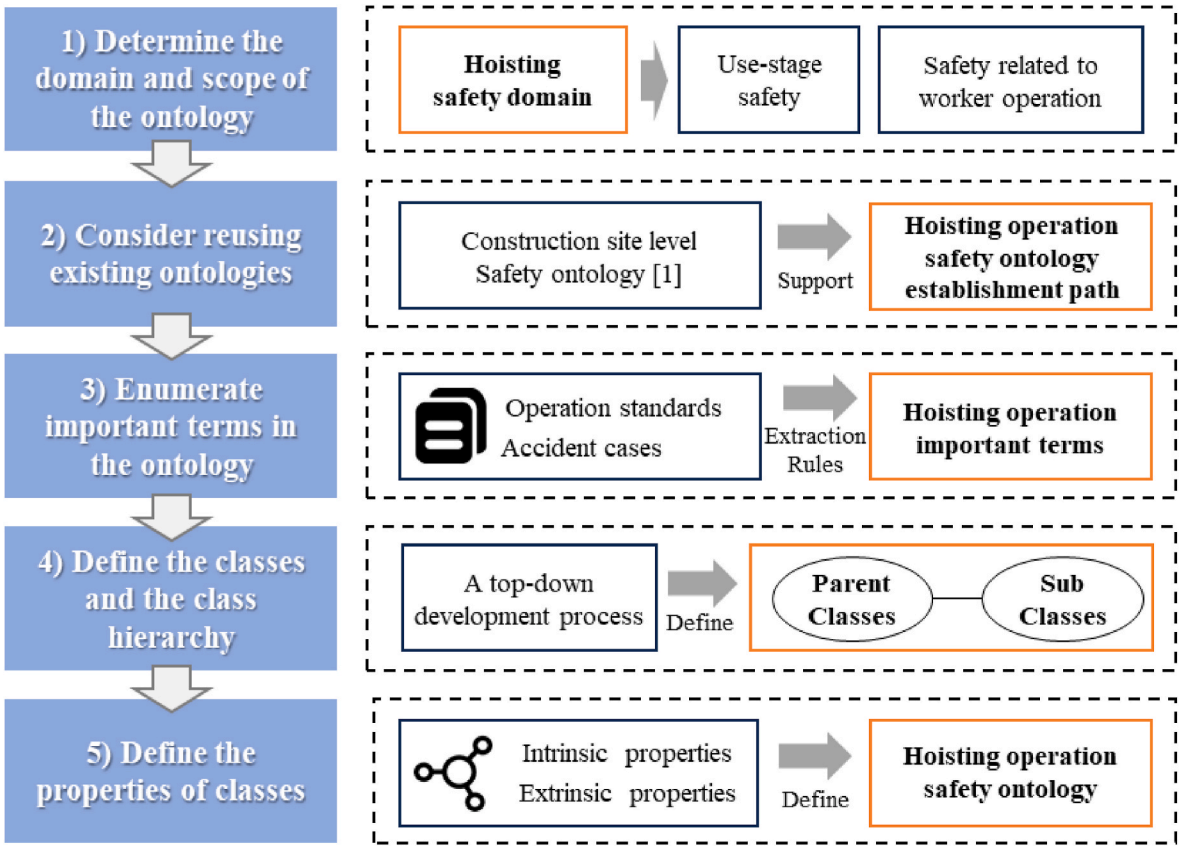


Fig. 3. Establishment process of hoisting operation ontology.

Table 2
Relevant standards.

No.	Standard
1	ANSI/ASME B30.3–2019 Tower Cranes
2	BS 7121-5-2019 Code of practice for safe use of cranes. Tower cranes
3	ISO 12480-3-2020 Cranes - Safe use - Part 3: Tower cranes
4	JISB 0146–3:2012 Cranes - Terminology - Part 3: Tower cranes
5	JGJ 196–2010 Technical specification for safety installation operation and dismantlement of tower crane in construction
6	JGJ 33–2012 Technical specification for safety operation of constructional machinery
7	GB/T 5082-2019 Crane-hand signals

oversight of unsafe hoisting behaviors during tower crane operations, underscoring the critical imperative for further investigative efforts.

3. Early-warning methodology for unsafe hoisting operations

3.1. DT of tower crane hoisting

The hoisting process represents a quintessential manifestation of a "Pick and Place" task, necessitating a synergistic collaboration among the driver, signalman, and rigger. Within the mechanical motion plane, hoisting entails three primary directions of motion: lifting, luffing, and rotation, often compounded by the rotational effect of the lifting rope. Each hoisting task is executed either autonomously or in conjunction with others. Concurrently, it is imperative to consider the influences stemming from wind conditions and the presence of surrounding obstacles.

Hence, in the endeavor to establish a DT model for tower crane hoisting, it is paramount to comprehensively integrate motion data of the inherent mechanisms' degrees of freedom and contextual environment data. The authors' prior research introduced the DT concept specifically tailored for mechanical motions of tower cranes (Jiang et al., 2022), underscoring the indispensability of integrating human-machine environment data through IoT. This integration facilitates functions such as mechanical simulation analysis and safety warnings through the interaction of virtual and physical information. Guided by these principles and the unique attributes of hoisting operations, the proposed DT for tower crane hoisting operations encompasses the following components.

(1) Formulating a virtual model that meticulously replicates the dimensional proportions and dynamic motion patterns of the tangible tower crane in the physical realm;
(2) Equipping the virtual tower crane with a well-defined methodology to identify unsafe hoisting operations.

This methodology is subsequently employed to ensure the operational safety of actual tower crane, as shown in Fig. 1.

The fundamental principle underpinning DT for tower crane hoisting lies in the mapping rules of the physical and virtual objects, as delineated in Fig. 2. This study, aligning with the operational sequence of tower crane activities, systematically categorizes the pertinent physical objects (including their material attribute) involved in hoisting operations, such as the jib, trolley, hook, rope, suspended load, and ambient conditions. These objects inherently give rise to distinct operational parameters, subsequently relayed to the virtual model through the mapping rule. Notably, the virtual object model encapsulates the representation of physical object, while the temporal evolution of attribute parameters is conveyed through time series data.

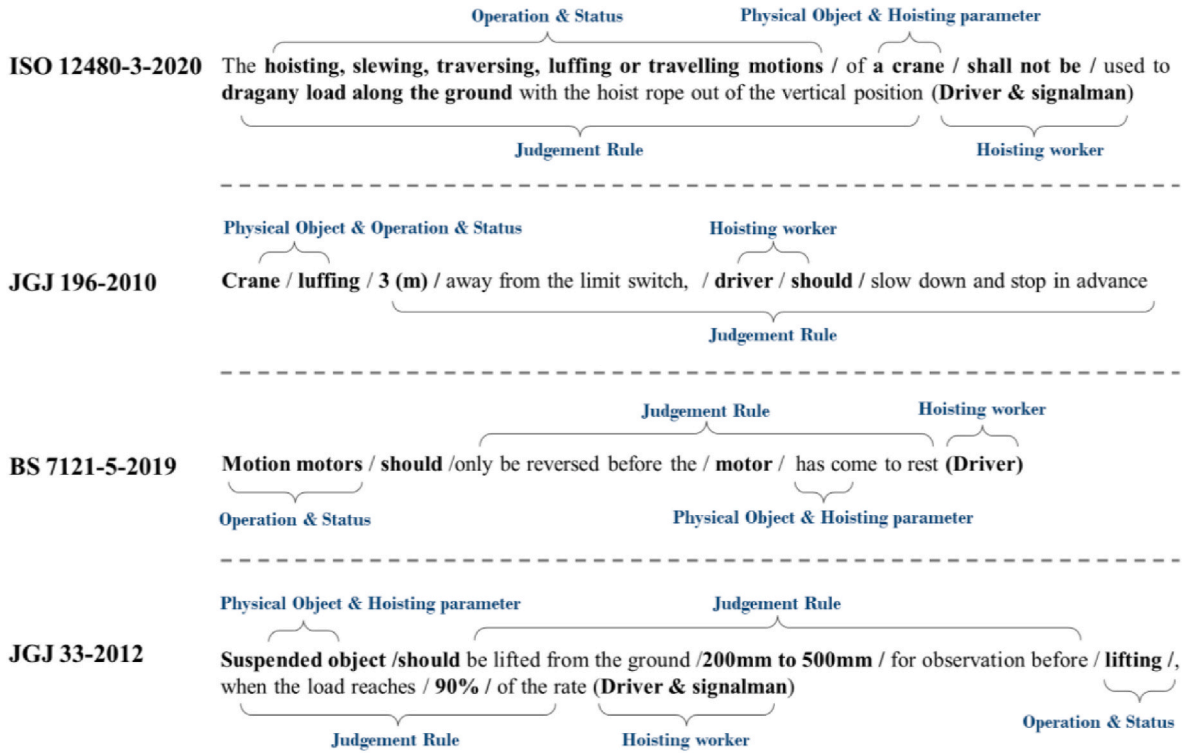


Fig. 4. Examples of standard texts.

Table 3
Class hierarchy.

No.	Parent class	Sub-class (ex.)	Description
1	Safety event	ex. Overload, overswing, untested lifting, tilt hoisting	Safety event caused by unsafe operations and status
2	Operation & Status	ex. Lifting, luffing, rotation, testing operation & motion	Hoisting workers' behavior & Physical objects' status
3	Judgement rule	ex. Swing angle > 5°, wind speed≥ 20 m/s, suspended weight > 0	Mathematical rule used to determine a lifting operation or status
4	Hoisting parameter	ex. Rotation angle, luffing distance, lifting height, wind speed	Parameter describing the status of an object
5	Physical object	ex. tower crane, wind, motor	Physical object in construction
6	Hoisting worker	ex. driver, signalman, rigger	Worker in construction

3.2. Knowledge graph for unsafe hoisting operations

3.2.1. Establishment step

Knowledge graph for unsafe hoisting operations serves as a domain ontology designed to standardize various entity knowledge concepts and their interrelationships. This ontology is pivotal in furnishing an early-warning capability for tower crane and its establishment draws upon established methods (Tang et al., 2023), as shown in Fig. 3.

1) **Scope definition:** The primary focus lies in delineating safety events stemming from unsafe hoisting operations. The ontology encompasses categories such as hoisting workers and related subjects, unsafe hoisting operations, safety events, and the relationship between them.
2) **Existing ontologies analysis:** The ontology on hoisting operations is limited, but safety ontologies have been cultivated in domains such as subway and building construction, drawing from accident case reports and regulatory standards (Xing et al., 2019)

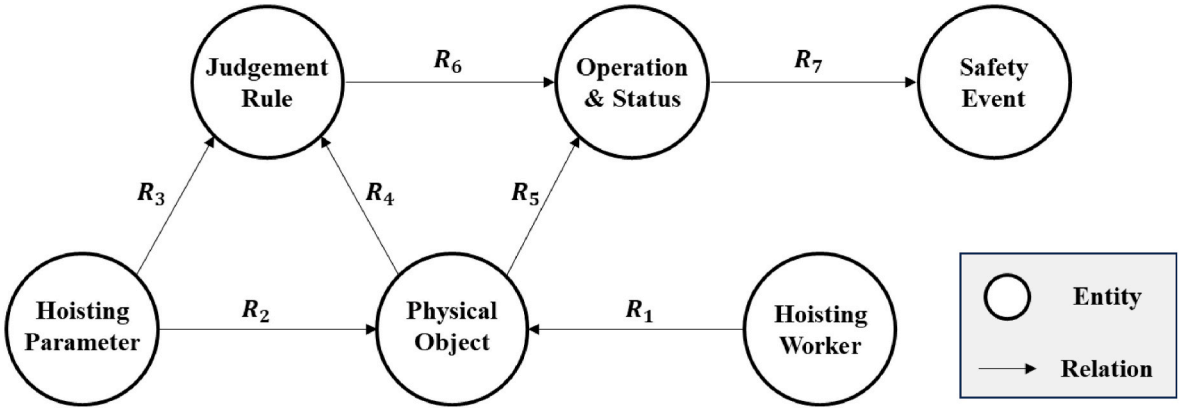


Fig. 5. Unsafe hoisting operation ontology.

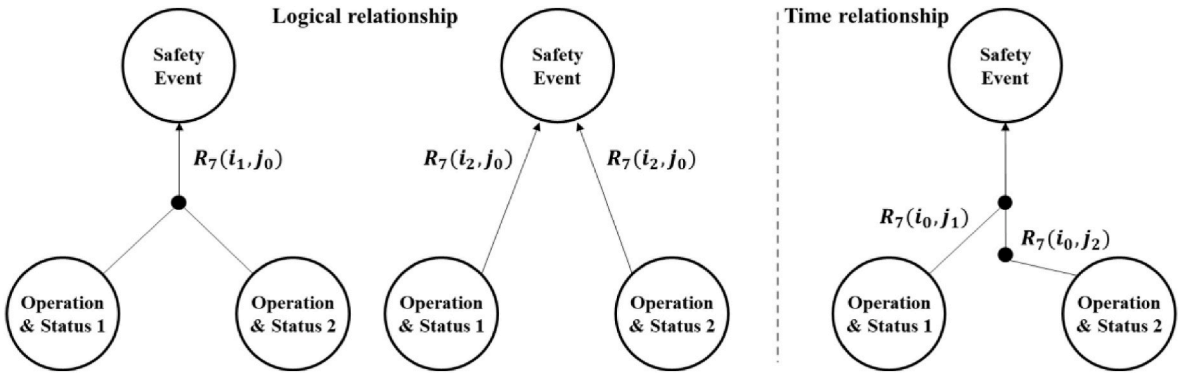


Fig. 6. Relationship between operation & status with safety event.

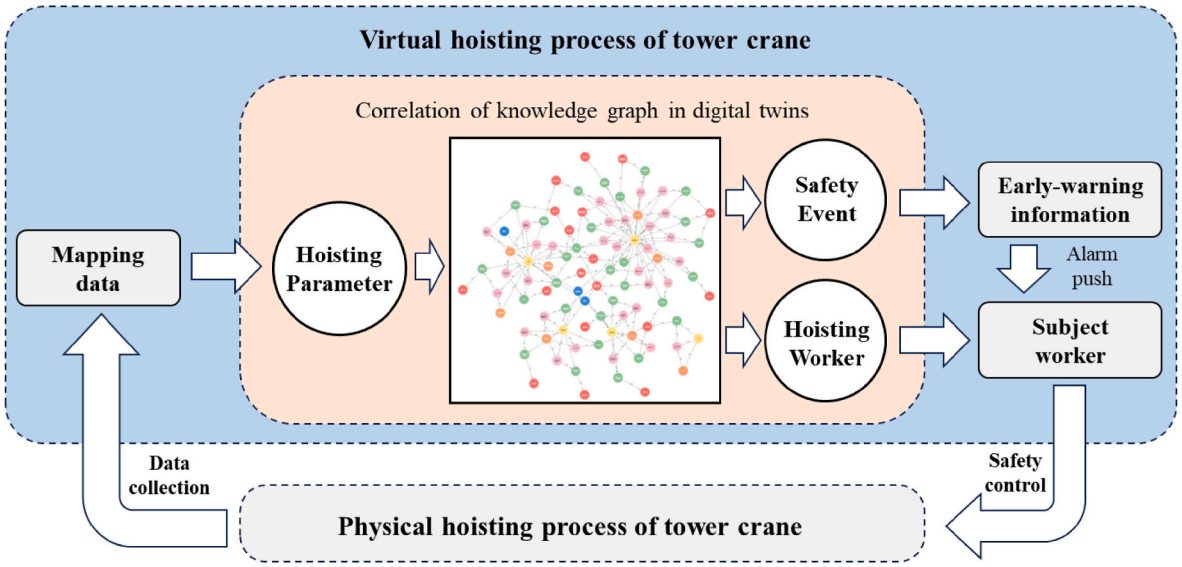


Fig. 7. Earth-warning process by DT and knowledge graph.



Fig. 8. Scaled tower crane and its control system.

(Gao et al., 2022). These serve as instructive precedents for refining the ontology concerning hoisting operations.

3) **Term description:** Terms are defined according to relevant standards, also focusing on: i) Use phase terms instead of organizational management, disassembly, and maintenance phases; ii) Hoisting worker terms instead of other workers on the construction site; iii) Worker's operation terms instead of their physical, psychological, and training conditions; iv) Inclusion of terms that can be easily monitored.

Table 4
Motion parameters of scaled tower crane.

Parameter	Type			
	Lifting	Luffing	Rotation	Load
Limit	0~1500 mm	0~700 mm	0~270°	5 kg
Speed	60 mm/s, 30 mm/s	30 mm/s, 10 mm/s	5°/s	/

4) **Class definition:** Classes serve as repositories for concepts sharing analogous properties and characteristics, facilitating the organization of related terminologies. We define six parent classes along with their respective subclasses, including safety events, hoisting operations or statuses, judgment rules, hoisting parameters, physical objects, and hoisting worker.

5) **Property specification** : Properties encompass both object and data types. Object-type properties denote relationships between two classes, articulated through a definition domain and a value domain. The ontology developed in this study concentrates on delineating concepts and relationships pertinent to safety events. Thus, object-type properties are predominantly employed, with further elaboration provided in Section 3.2.2.

This study establishes a hoisting operation ontology primarily informed by crane standards. Seventeen contemporary domestic and international standards underwent thorough analysis, culminating in the selection of seven highly pertinent and representative standards, as delineated in Table 2. Furthermore, recognizing the succinct nature of the standards, the study also compiles 86 crane accident case reports released by Chinese government agencies spanning from 2010 to 2023. The analysis of these accidents serves to complement the ontology of unsafe hoisting operations (Jiang, 2022).

3.2.2. *Term description and class definition*

The definition of terms and classes is established through a synthesis of domain scope and relevant standards. From the domain scope perspective, this includes terms such as hoisting workers and operational status. By integrating standards, the class is expanded to encompass entities such as hoisting workers, physical objects, judgment rules, operation & status, hoisting parameters, and safety events. Fig. 4 illustrates examples of specific standard texts, revealing that not every text directly represents all entities within the class category. Therefore, incomplete texts can be supplemented based on expert experience. For instance, the hoisting worker entity can be supplemented through the association of operational activities, while hoisting parameters, as vital components of the DT in this study, can be inferred and analyzed through the association of physical object entity. Furthermore, judgement rules are described using words like “should” and “not” as indicators. Additionally, due to inconsistent vocabulary across standards,

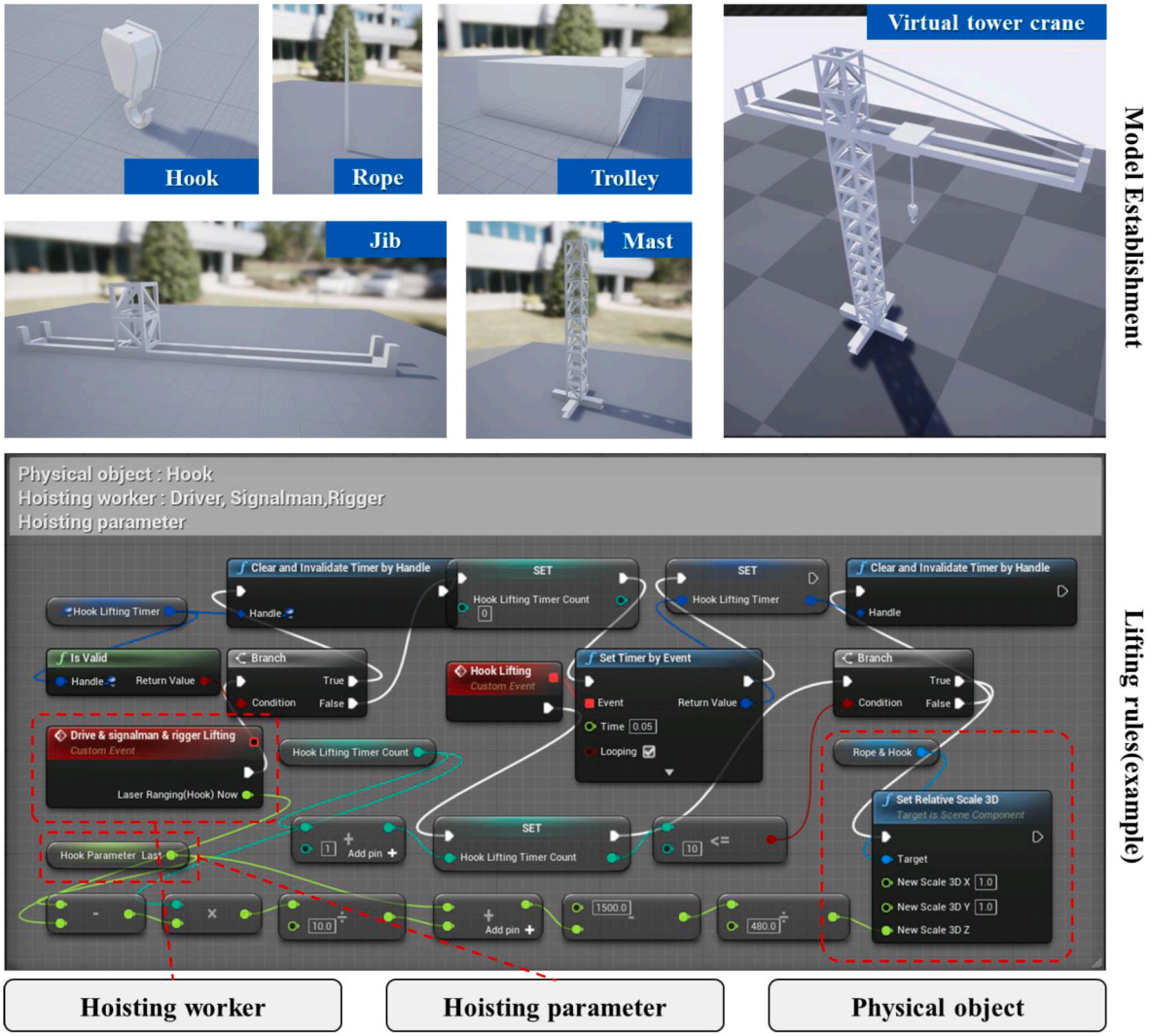


Fig. 9. Establishment of the virtual tower crane.

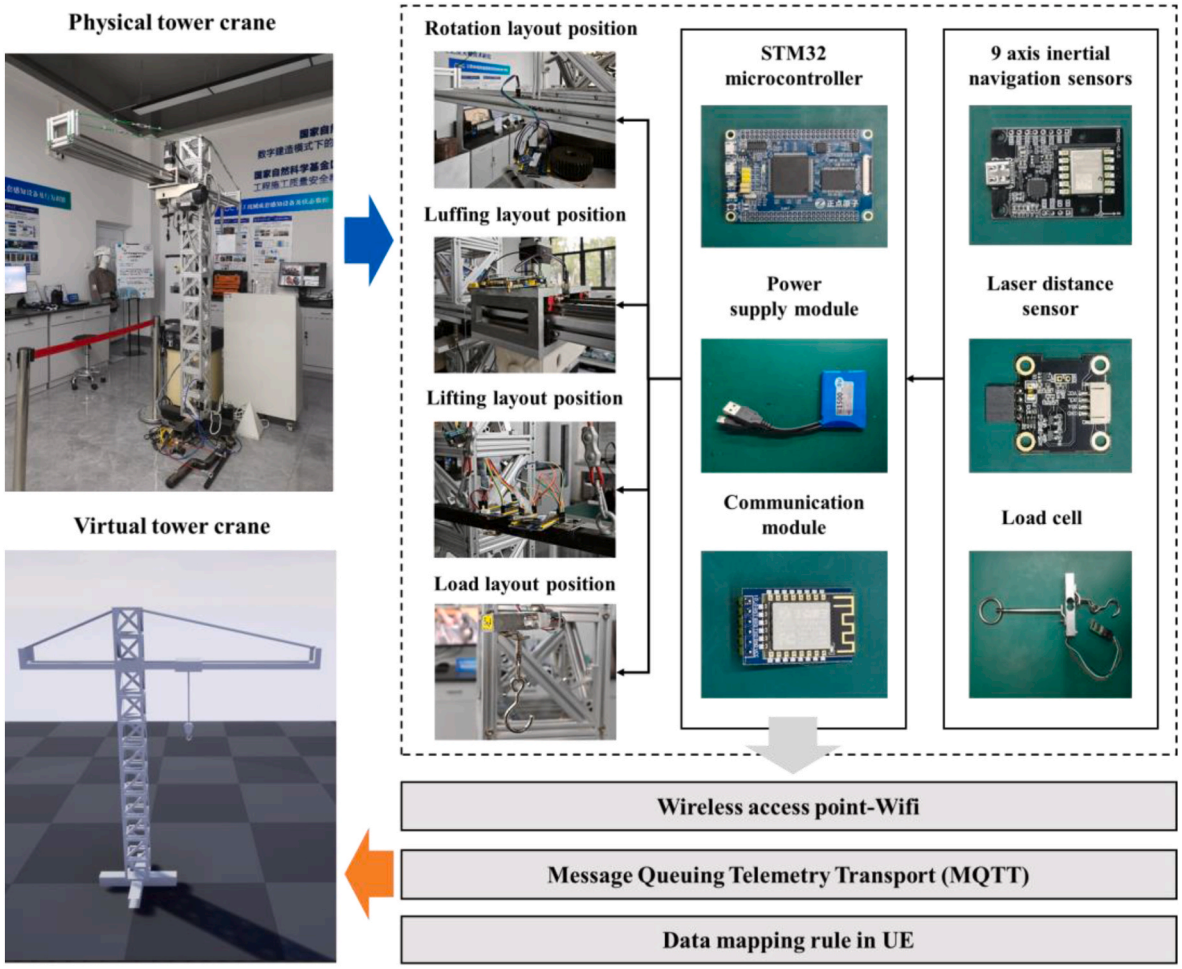


Fig. 10. Sensors and hoisting data mapping.

synonymous terms from various standards are compiled to support the development of the knowledge graph, as outlined in Table 3.

3.2.3. *Unsafe hoisting operation ontology*

The ontology depicting unsafe hoisting behaviors, as delineated in Fig. 5 through triple form, is established via the interpretation of standard specifications. At the semantic level, several distinct relations are identified: R1 denotes the operational relation, R2 the feedback relation, R3 and R4 the foundational relations, R5 the media relation, R6 the decision relation, and R7 the causal relation. Notably, R1-R6 exhibit one-to-one correspondence.

The relational aspect represented by R7 is intricate, where diverse combinations of hoisting operations and statuses can engender one or more safety events. Consequently, $R_7(i,j)$ is established to describe the relationship between different operations/statuses and safety events. Herein, i is for logical relation, j is for time relation, as shown in Fig. 6. The definition of i is specified as $i = (i_0, i_1, i_2)$, wherein i_0 denotes the absence of any association between safety events and operations/statuses; i_1 denotes a conjunctive relationship "AND" between safety events and operations/statuses; i_2 indicates a disjunctive relationship "OR" between safety events and operations/statuses. Similarly j is defined as $j = (j_0, j_1, j_2, \ldots\ldots j_n)$, j_0 indicates the absence of time sequence relationship between safety events and operations/statuses; j_1 signifies the precedence of operations/statuses over others during safety events. A larger n value indicates a greater temporal distance.

3.3. *Integrating DT and knowledge graph for early-warning*

The integration of DT with knowledge graph represents a crucial step towards achieving data-driven early-warning of unsafe hoisting operations, as depicted in Fig. 7. This association mechanism is multifaceted: Firstly, the mapping data is linked with hoisting parameters within the knowledge graph, serving as input. Secondly, according to hoisting parameters, the knowledge graph outputs safety event entities. In turn, the early-warning information is visualized within the DT and disseminated to pertinent workers, thereby facilitating comprehensive control feedback.

4. Validation through laboratory experiment

4.1. *Scaled tower crane and its digital twin*

This study adopts an experimental approach to validated the proposed early-warning methodology, utilizing a scaled replica of a tower crane. In our previous research effort, a meticulously crafted scaled model was developed, maintaining an approximate geometric ratio of "1:80". This scaled-down model underwent comprehensive assessment, confirming its ability to replicate certain characteristics of a full-scale tower crane in terms of motion patterns, boundary constraints, and material property distribution (Jiang et al., 2022), as illustrated in Fig. 8.

Functionally, the motion system of the scaled tower crane model comprises three primary mechanisms: the lifting mechanism, rotation mechanism, and luffing mechanism, each with specified boundary

Table 5
Knowledge graph structure of unsafe hoisting operation.

Entity	Safety Event	$R_7(i,j)$	Operation & Status	Judgement rule	Hoisting parameter	Physical object	Hoisting worker
SE1	Suspended in the air for a long time	$R_7(i_1,j_0)$	Suspended status	SSW > 0	SSW	Suspended	Driver
			Lifting operation	HLH > 0	HLS, HLH	Hook	Signalman
				HLS = 0, T_i			
			Luffing operation	TLS = 0, T_i	TLS	Trolley	
SE2	Suspended is scattered	$R_7(i_1,j_1)$	Lifting operation	HLH > 0	HLH	Hook	Rigger
			Suspended status	SSW > 0	SSW	Suspended	
		$R_7(i_1,j_2)$	Suspended status	SSW_1- SSW_2 > 0	SSW	Suspended	
SE3	Untested Lifting	$R_7(i_1,j_0)$	Rotation operation	JRS = 0, T_i	JRS	Jib	Driver Signalman
			Lifting operation	HLH < HLH_0	HLH, HLS	Hook	
				HLS > 0			
SE4	Tilt hoisting	$R_7(i_1,j_1)$	Lifting operation	HLH = 0	HLH	Hook	Driver
		$R_7(i_1,j_2)$	Lifting operation	HLH > 0	HLH	Hook	Signalman
			Suspended status	HW > 0	HW, SSA	Suspended	
				SSA > SSA_0			
SE5	Reverse operation without stopping in advance	$R_7(i_2,j_0)$	Lifting operation	HLS_1*HLS_2 < 0	HLS	Hook	Driver
			Luffing operation	TLS_1*TLS_2 < 0	TLS	Trolley	
			Rotation operation	JRS_1*JRS_2 < 0	JRS	Jib	
SE6	Sudden acceleration operation	$R_7(i_2,j_0)$	Lifting operation	HLS_1-HLS_2 > *HL* S_0	HLS	Hook	Driver
			Luffing operation	TLS_1-TLS_2 > *TL* S_0	TLS	Trolley	
			Rotation operation	JRS_1-JRS_2 > JRS_0	JRS	Jib	
SE7	Suspended over swings or tilts	$R_7(i_1,j_2)$	Suspended status	SSW > 0	SSW	Suspended	Rigger
		$R_7(i_2,j_0)$	Suspended status	SSA > SSA_0	SSA	Suspended	
			Suspended status	SIA > SIA_0	SIA	Suspended	
SE8	Hoisting in strong wind	$R_7(i_1,j_0)$	Environment status	EWS > EWS_0	EWS	Environment	Driver Signalman
			Lifting operation	HLS > 0	HLS	Hook	
			Luffing operation	TLS > 0	TLS	Trolley	
			Rotation operation	JRS > 0	JRS	Jib	

Annotation.
JRA-Rotation angel; JRS-Rotation speed; TLD-Luffing distance; TLS-Luffing speed; HLH-Lifting height.
HLS-Lifting speed; SSA-Suspended swing angle; SIA- Suspended inclination angle; SSW- Suspended self-weight.
EWS-Wind speed; T_i- Critical time; Subscript [Number 1 and Number 2]- Two consecutive values; Subscript [Number 0]- Critical value.

parameters outlined in Table 4. The lifting mechanism utilizes an electric hoist with two lifting speeds: 60 mm/s and 30 mm/s. Meanwhile, the rotation and luffing mechanisms employ the Mitsubishi servo motor KN43J-S100, enabling a maximum rotational speed of 3000 r/min and supporting programmable logic control. The control system governing the scaled tower crane operates in two modes: automatic control and manual control. The automatic mode enables precise positioning by quantitatively inputting the luffing distance and rotation angle. Conversely, the manual mode replicates real-world crane operation procedures, facilitating the simulation of various unsafe hoisting operations and enabling a comprehensive study of potential safety concerns.

The DT of the tower crane comprises two principal components: formulating the virtual model for the tower crane structure and its corresponding motion regulations, and integrating the proposed graph model that identifies unsafe hoisting operations. In this study, a precise virtual model of this scaled tower crane was crafted using SolidWorks software. This intricate model contains vital components including the hoist, mast, jib, counter-jib, ropes, as well as intricate motion mechanisms like the hook and luffing apparatus. Subsequently, the entire model was imported into the Unity Engine (UE) platform, facilitating the establishment of mapping rules governing the tower crane's range of motion. These rules governed degrees of freedom related to lifting, luffing, rotation, and hook swinging, while simultaneously addressing local constraints concerning the crane's foundation, structural elements, and intricate mechanical interconnections, as depicted in Fig. 9.

This study adopted a third-party service approach to oversee and monitor unsafe hoisting operations. A data collection method relying on attachment sensors was implemented during experiments, involving the establishment of data collection infrastructure and the formulation of transmission protocols. Multiple STM32 microcontrollers interface power modules, communication modules, and sensor modules, incorporating three 9-axis IMU sensors and two laser distance sensors. Positioned strategically on the lifting, luffing, and rotation mechanisms, IMU sensors measure rotation angles, luffing speeds, lifting speeds, and swinging angles. Magnetic adhesion connections between sensors and various mechanisms ensure flexibility for diverse hoisting tasks.

Communication between sensors and the virtual tower crane is facilitated by a wireless architecture rooted in the Message Queuing Telemetry Transport (MQTT) protocol. STM32 microcontrollers initiate connections with nearby wireless access points, subsequently linking to the MQTT server for data transmission. On the user end, an MQTT communication module retrieves sensor data from the server, capturing transmitted readings. This data is correlated with the degrees of freedom exhibited by various components within the virtual model, ensuring seamless synchronization with the actual movements of the tower crane, as illustrated in Fig. 10.

4.2. Knowledge graph generation and correlation

4.2.1. Structure of the knowledge graph

Following the methodology outlined, the resultant structure of the knowledge graph is shown in Table 5. It is worth noting that the knowledge graph can be further enriched as additional standards and reports become available. Based on the Neo4j-admin import tool, the knowledge graph is visualized using the Load.csv instruction, as exemplified in Fig. 11. In this regard, we employ the knowledge graph of SE6 as an example and validate its logical validity by comparing it with the pertinent standards.

4.2.2. Knowledge graph implementation

To implement the knowledge graph in the DT for early warning, entities pertinent to unsafe hoisting operations are extracted and formalized into an ontology. Subsequently, these entities and their interrelations are translated into computationally comprehensible logical statements, leading to the formulation of early warning rules in UE. Notably, distinct node information is defined for various entities in UE, with particular emphasis on representing hoisting parameters and judgment rules as byte-level data (i.e., numeric or string value).

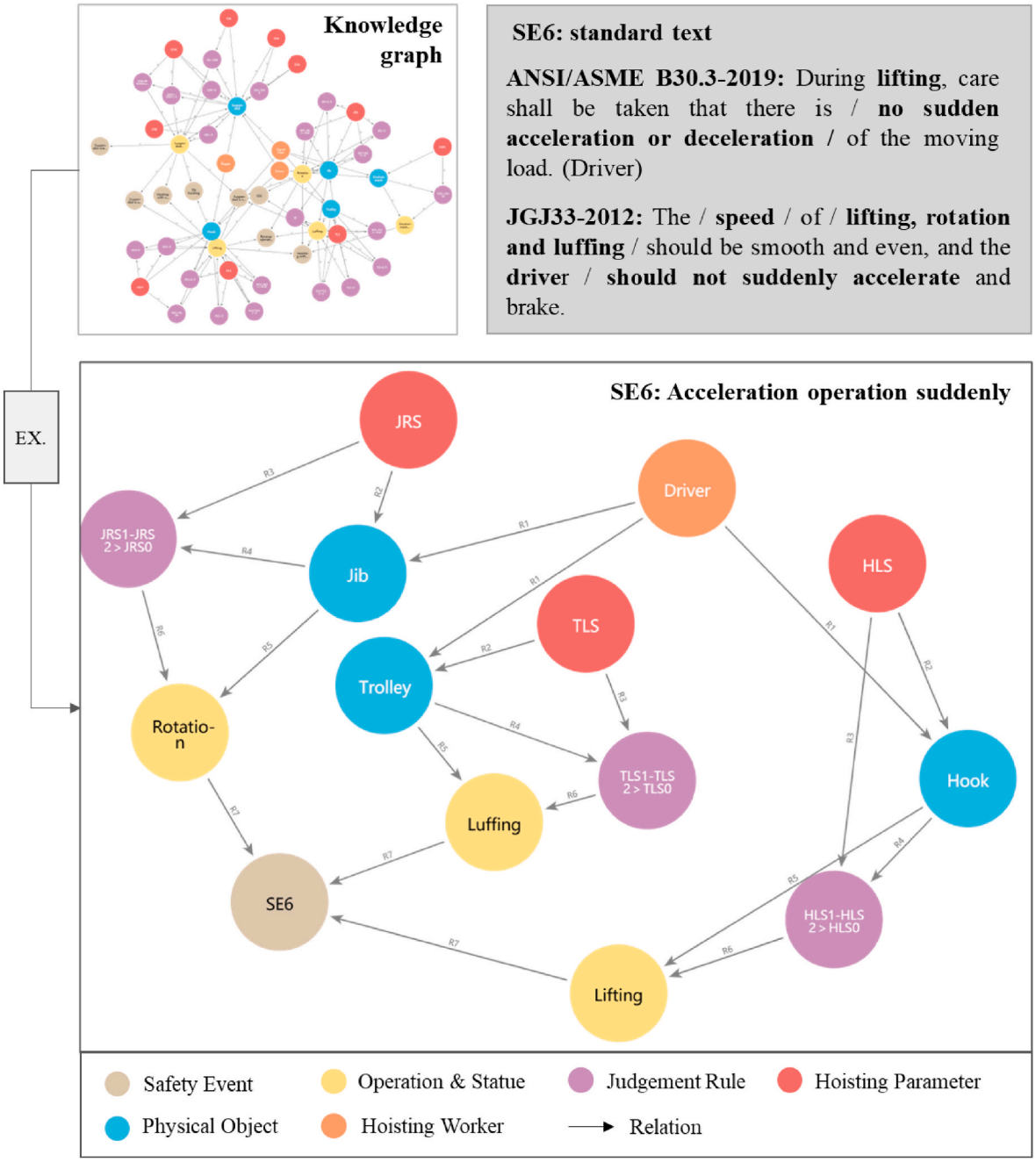


Fig. 11. Knowledge graph of unsafe hoisting operation.

Progressive relationships between entities are then established, categorized into two main types: those reliant on Boolean judgments and those guided by predefined rules in the judgment rule entity. Following this, the input corresponding to the DT data should be associated with hoisting parameters, while the output encompasses information regarding safety events and involved hoisting workers.

An example of the SE6 knowledge graph in the DT is depicted in Fig. 12. During the development of the DT (see Fig. 9), entities such as hoisting parameter, physical object, and hoisting worker have been established. Therefore, it is only necessary to associate relevant entity information for hoisting parameters. Simultaneously, the creation of decision rule entities, operation & status entities, and the association of safety events, along with integration with the user interface and the early-warning output, enables the application of the knowledge graph.

Subsequently, data synchronization assumes pivotal importance in promptly identifying unsafe hoisting operations. An early-warning system interface is developed, as depicted in Fig. 13, which shows the hoisting workers information, safety event information and hoisting data information by the associated with knowledge graph. Upon detection of unsafe hoisting operations, safety managers receive timely notifications via warning messages, facilitating prompt corrective actions.

4.3. Hoisting experiment of the scaled tower crane

To validate the accuracy of DT mapping for the scaled tower crane and its efficacy in warning unsafe hoisting operations, a two-stage hoisting experiment was conducted. The experimental design mainly considers two principles: i) motion parameters cover three motion modes; ii) setting more than two path parameters for each mode of

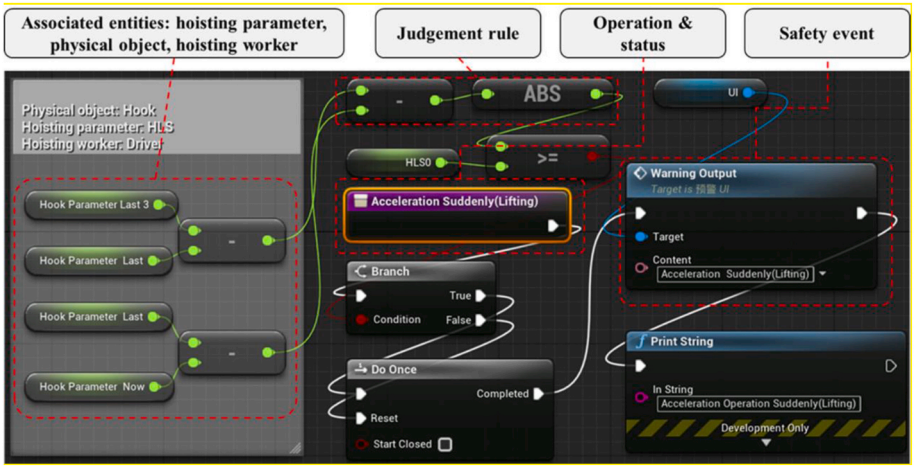


Fig. 12. Example of SE6 knowledge graph in the DT.

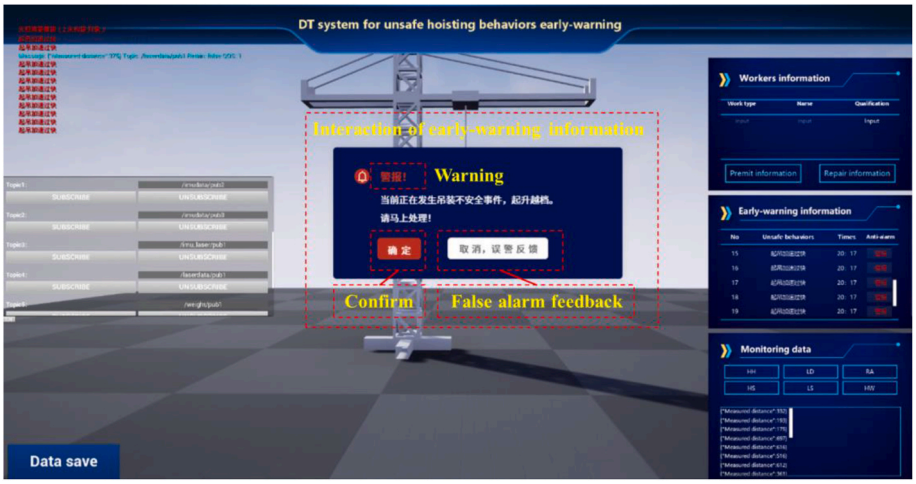


Fig. 13. Early-warning system interface.

Table 6
Experiment parameters of different motions.

No.	Motion	Initial attitude → Termination attitude (Lift/mm, Luff/mm, Rotate/°)	Load
1	Lifting	(0,300,0)→(530,300,0)→(0,300,0)	2 kg
2		(0,300,0)→(900,300,0)→(0,300,0)	2 kg
3		(0,300,0)→(1140,300,0)→(0,300,0)	2 kg
5	Luffing	(300,200,0)→(300,300,0)→(300,200,0)	2 kg
6		(300,200,0)→(300,400,0)→(300,200,0)	2 kg
7		(300,200,0)→(300,450,0)→(300,200,0)	2 kg
10	Rotation	(300,300,0)→(300,300,-20)→(300,300,0)	2 kg
11		(300,300,0)→(300,300,-30)→(300,300,0)	2 kg
12		(300,300,0)→(300,300,-40)→(300,300,0)	2 kg

motion and each safety event. In the initial stage, distinct motion experiments were undertaken to assess the precision of motion data collection and mapping for lifting, luffing, and rotation motions of the tower crane, guided by the parameters outlined in Table 6. Subsequently, the second stage involved a series of experiments encompassing various forms of unsafe hoisting operations detailed in Table 7 (excluding SE8, due to unavailable simulation conditions). Each experiment was documented twice, initially focusing on the response to instances of unsafe hoisting triggering early warnings, followed by observation of the ceasing warnings upon rectification of the unsafe

Table 7
Experiment parameters of unsafe hoisting operations.

No.	Safety Event	Initial attitude → Termination attitude (Lift/mm, Luff/mm, Rotate/°)	Speed	Times	Load
1	SE1	(300, 200, 0)	0	30s	2 kg
2		(300, 200, 0)→(400, 200, 0)	30 mm/s	/	2 kg
4	SE2	(600, 450, 0)→(600, 650, 0)	10 mm/s	/	2 kg
5		(600, 650, 0)→(600, 450, 0)	10 mm/s	/	2 kg
7	SE3	(0, 400, 0)→(600, 400, 0)	30 mm/s	/	4.5 kg
8		(0, 400, 0)→(600, 400, 0)	30 mm/s	/	2 kg
10	SE4	(600, 700, 0)	10 mm/s	30s	2 kg
11		(600, 700, 0)→(600, 400, 0)	10 mm/s	/	2 kg
13	SE5	(640, 200, 0)→(640, 200, 30)→(640, 200, 20)	5°/s	/	2 kg
14		(640, 200, 0)→(640, 200, 30)	5°/s	/	2 kg
15	SE6	(0, 200, 0)→(600, 200, 0)	0 → 60 mm/s	/	2 kg
16		(0, 200, 0)→(600, 200, 0)	0 → 30 mm/s	/	2 kg
17	SE7	(0, 200, 0)→(400, 200, 0), > 5° swing	30 mm/s	/	2 kg
18		(0, 200, 0)→(400, 200, 0), < 5° swing	30 mm/s	/	2 kg

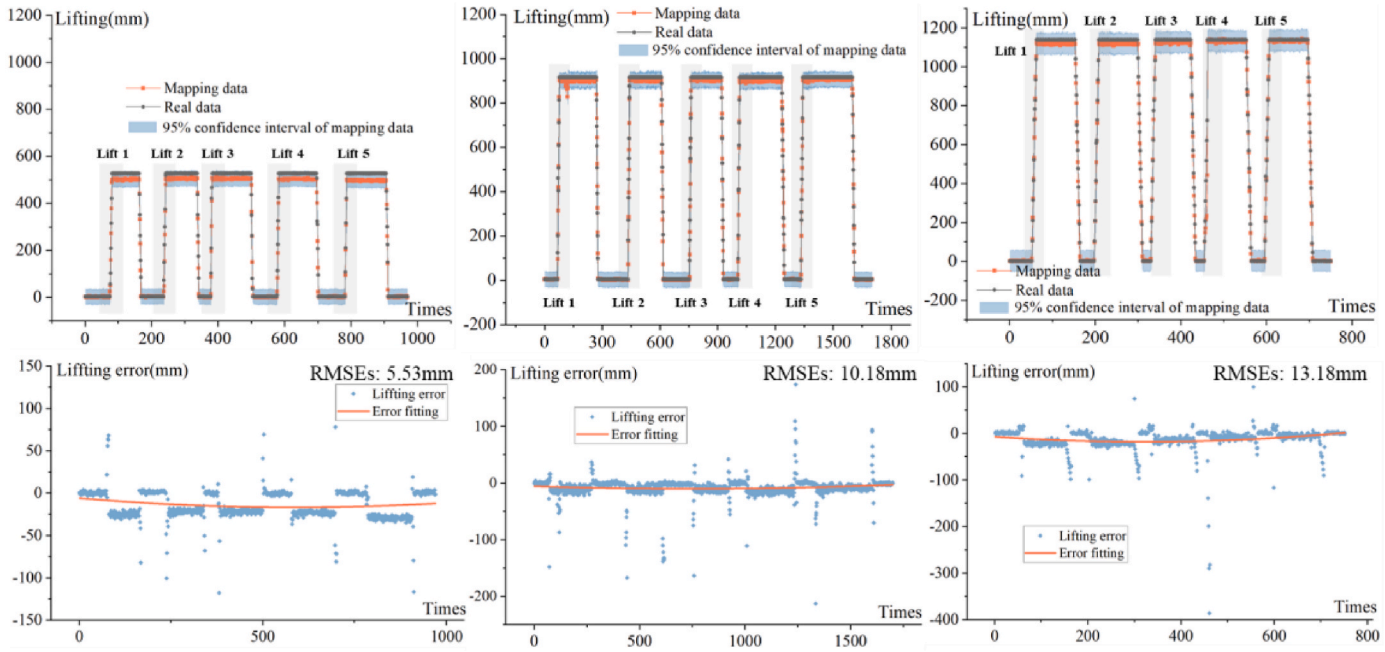


Fig. 14. Mapping accuracy of lifting motion.

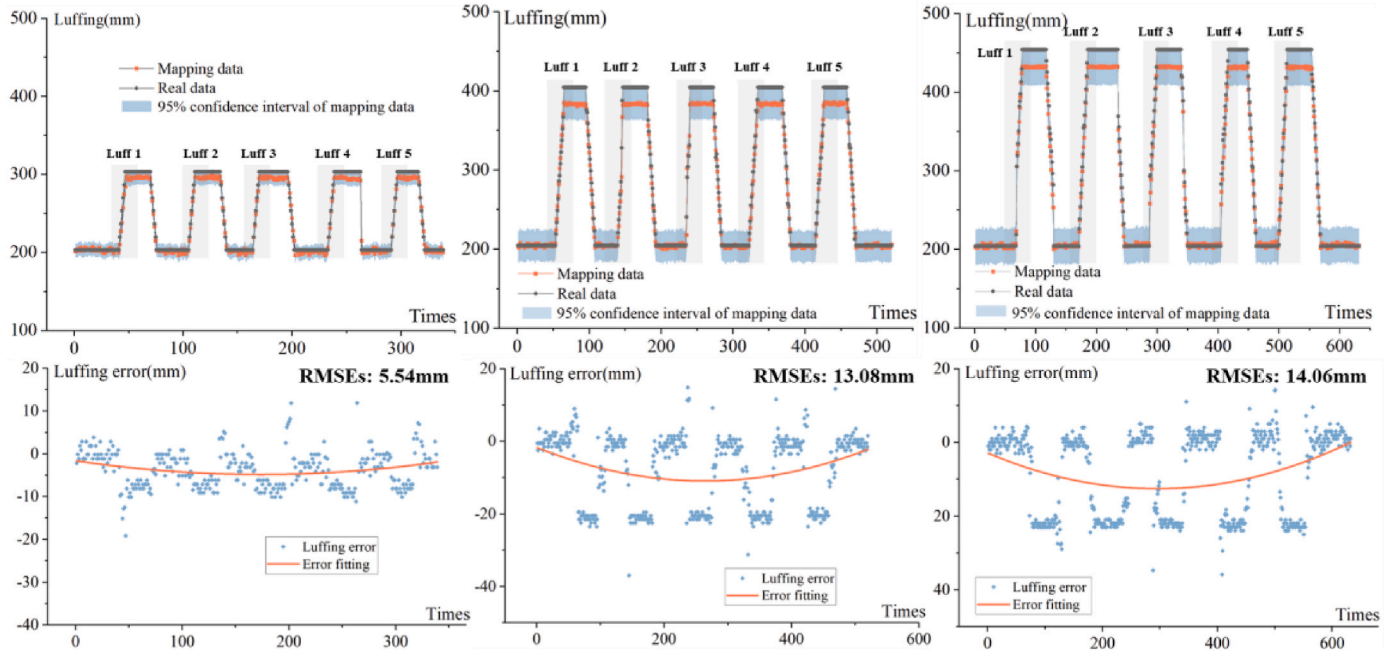


Fig. 15. Mapping accuracy of luffing motion.

hoisting operation. This approach facilitated assessment of both the early warning rate and its rate of false alarms, with each experiment type replicated ten times to ensure robustness of the results.

5. Results and discussions

5.1. Analysis of mapping accuracy

The precision of data mapping for hoisting, luffing, and rotation was evaluated based on experimental outcomes, as illustrated in Figs. 14–16, respectively. Each figure presents actual data (depicted in black), mapped data (highlighted in orange), and a 95% confidence interval error band (indicated by the blue band). On the right side, error variance and quadratic fitting curves are displayed.

Regarding lifting data (see Fig. 14), theoretical data for three sets of lifting motions predominantly fell within the confidence interval of mapped data, affirming data collection accuracy. Consistency in lifting errors across experiments suggests stable variations for each hoisting instance. Fitted curves indicate absolute errors within a 25 mm range across all three experiment sets, with fluctuations attributed to power system influences at hoisting initiation and termination, and the RMSEs of lifting are 5.53 mm, 10.18 mm, 13.18 mm. Nevertheless, on the whole, mapped lifting data can effectively identify unsafe hoisting operations.

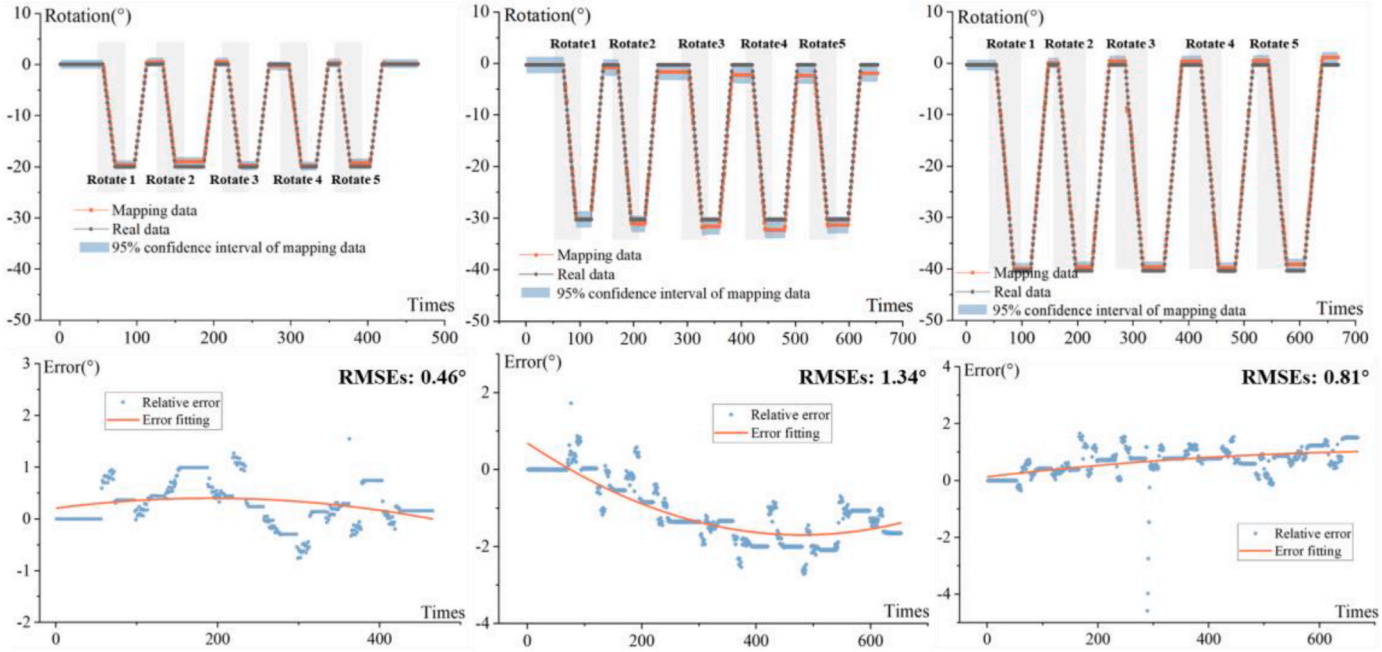


Fig. 16. Mapping accuracy of rotation motion.

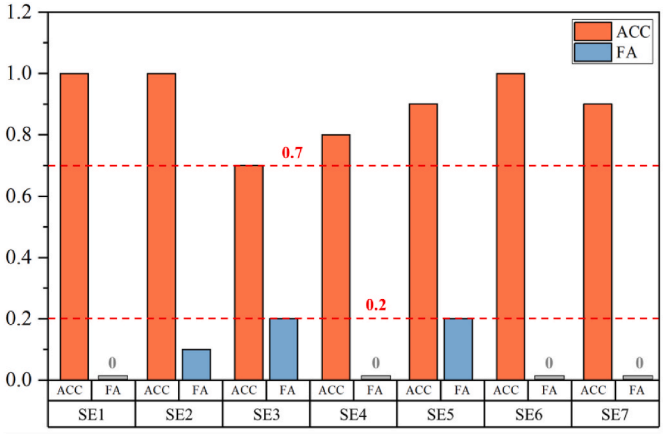


Fig. 17. Early-warning accuracy of different unsafe hoisting operations.

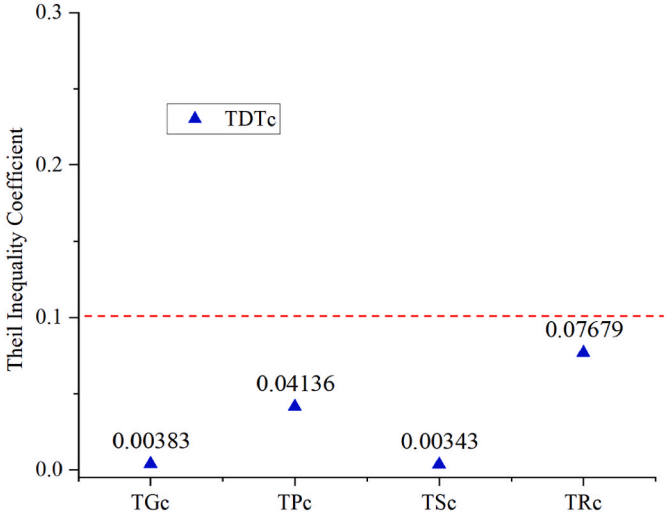


Fig. 18. Consistency analysis of DT.

In luffing data analysis (see Fig. 15), theoretical data for three sets of luffing motions also fell within the confidence interval of mapped data. Consistent stability in experimental errors across trials was observed. Fitted curves revealed absolute errors within 10 mm for smaller luffing distances and within 20 mm for larger ones, and the RMSEs of luffing are 5.54 mm, 13.08 mm, 14.06 mm. This observation underscores the minimal cumulative error in mapped luffing data, enhancing the ability to gather long-term operational data.

Rotation data analysis (see Fig. 16) showed theoretical data for three sets of rotation motions within the confidence interval of mapped data. Within the same set of rotation maneuvers, a gradual escalation of error was noted over five consecutive experiments, and the RMSEs of rotation are 0.46°, 1.34°, 0.81°. This tendency is further highlighted in the fitting curve of errors, where a degree of cumulative error becomes evident in larger rotation movements (30°, 40°), but the overall error remained within 2°.

Overall, the constructed DT accurately maps tower crane operational postures, aiding in identifying unsafe hoisting operations effectively.

5.2. Analysis of early-warning accuracy

This study extensively analyzed both ACC (Accuracy) and FA (False Alarm) indices for early-warning of unsafe hoisting operations. ACC denotes the precision in generating alerts during unsafe hoisting, calculated as the ratio of accurate identifications to total experiments; FA represents the rate of false alarms during routine hoisting operations under consistent conditions, calculated as the ratio of incorrect identifications to total experiments. The findings presented in Fig. 17 are based on the experimental setup detailed in Table 7. Each set of data was obtained from the results of ten repeated experiments.

Experiments revealed consistent ACC of 0.7 across all operation types, with SE1, SE2, SE6, and SE7 showing higher ACC and low FA. However, SE3 exhibited low ACC and high FA due to fluctuating weight thresholds during lifting acceleration and deceleration, along with variations in test hoist height distances from actual operation scenarios. Moreover, SE5 (reverse operation) demonstrated elevated false alarm

potential. This outcome primarily stemmed from pronounced reverse acceleration upon rotation halting at high speeds, leading to erroneous alerts during reverse turning scenarios.

Furthermore, false alarms may be influenced by factors such as mapping accuracy and system errors. Therefore, to ensure the efficacy of the early-warning system in real-world construction contexts, this study posits that the mechanism for identifying unsafe hoisting operations should incorporate both the early-warning system and manual verification. As depicted in Fig. 13, personnel receiving interface alerts can collaborate with on-site supervisors to validate the final outcomes. Conversely, the knowledge graph links safety events with hoisting workers. Consequently, the early-warning system can conduct a comprehensive assessment of different workers. This approach mitigates the risk of erroneous evaluations stemming from minor false alarm occurrences.

5.3. Consistency analysis for DT of tower crane

The evaluation of DT consistency holds significant importance as it serves as a primary mechanism to assess the practical applicability of DT. In this discourse, we delve into the assessment of the virtual-physical consistency of the developed DT of the tower crane, drawing upon established methodologies (Zhang et al., 2022) (Uhlenkamp et al., 2022). This evaluation primarily encompasses four key indices: geometric dimension, physical attribute, operational status, and regulatory protocols. Geometric dimension consistency scrutinizes the variance between the structural geometric dimensions and their numerical representations. Physical attributes predominantly pertain to mechanical properties, gauged through dynamic response data of the tower crane structure. Operational status reflects the hoisting activities, assessed by the precision of lifting, luffing, and rotation variation. Regulatory compliance is evaluated by the warning accuracy pertaining to operational safety incidents. For these indices, comprehensive numerical data, inclusive of geometric dimensions and physical properties, have been compiled from our prior research (Jiang et al., 2022), and operational status and regulatory adherence are captured as depicted in Figs. 14–17. Given the array-like and continuous nature of each index, the Theil Inequality Coefficient (TIC) (Roungas et al., 2018) is used to evaluate the DT consistency, as shown in Formula (1).

$$\begin{cases} TDT_c \sim \{TG_c, TP_c, TS_c, TR_c\} \\ TDT_c = \dfrac{\sqrt{\dfrac{1}{N}\sum_{i=1}^{N}(x_i - y_i)^2}}{\sqrt{\dfrac{1}{N}\sum_{i=1}^{N}x_i^2} + \sqrt{\dfrac{1}{N}\sum_{i=1}^{N}y_i^2}} \end{cases} \quad \text{Formula (1)}$$

Where TDT_c refers to the consistency index for DT of tower crane; TG_c is the geometric dimension consistency index of; TP_c is the physical attribute consistency index; TS_c is the operational status consistency index; TR_c is the regulatory compliance consistency index; x_i is the real sample; y_i is the virtual sample; N denotes the sample number.

The computed consistency results are presented in Fig. 18. Statistically, a TIC value below 0.1 signifies superior consistency, a criterion satisfactorily met by each index of TDT_c. This observation indicates the commendable practical utility of the developed DT of the tower crane. Notably, TG_c and TS_c exhibit superior consistency performance, suggesting the feasibility of achieving virtual-to-real mappings for geometric dimension and operational status. Conversely, TP_c and TR_c manifest slightly weaker consistency, underscoring challenges in virtual-to-real mappings of physical attribute and regulatory compliance, thus warranting further research focus.

6. Conclusion

The study introduces an early-warning methodology that combines DT and knowledge graph to promptly identify unsafe hoisting operations of tower cranes. A pivotal aspect of this methodology involves synchronously mapping operational data for the tower crane by creating a DT. Subsequently, this data undergoes analysis to detect unsafe hoisting operations using a bespoke knowledge graph. The effectiveness of this methodology is validated through meticulously designed experiments, wherein the results indicate that the confidence intervals of the mapped data pertaining to lifting, luffing, and rotation variations encompass the theoretical data. The results also reveal that the warning rate for identifying unsafe hoisting operations surpass 70%, while simultaneously maintaining the false alarm rate below 20%. Moreover, the DT consistency is evaluated, and high-quality mapping is realized in geometric size, physical properties, operation status and management rules. These findings underscore the efficacy of the proposed methodology in pinpointing and preventing unsafe hoisting operations.

Nonetheless, it is important to acknowledge that this study primarily focused on a scaled-down tower crane within a laboratory setting, neglecting the influence of the dynamic site environment on the precision of the warnings. In more intricate on-site operations, there likely exist operator-specific habits, which introduce further disturbances that necessitate additional investigation. Additionally, the introduced DT may encounter cumulative errors in terms of data mapping.

Future research directions can be pursued along two avenues. Firstly, there is a need to analyze the sources of consistency disparities in the DT of the tower crane and to develop a more comprehensive mapping framework. Secondly, it is essential to expand the range of operational scenarios for tower cranes by accruing empirical data from real-world hoisting activities.

CRediT authorship contribution statement

Weiguang Jiang: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Yuhan Liu:** Writing – original draft, Validation, Investigation, Formal analysis. **Ke Chen:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Yihong Liu:** Investigation, Formal analysis. **Lieyun Ding:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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